

STAT 312: HW Assignment 3 - Unit 04 & 05

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Submission Guidelines

Please follow these guidelines carefully when completing and submitting your homework. Clear and well-organized work helps ensure fair and accurate grading.

- There are **7 problems**.
- Submit all homework assignments through **Canvas**.
- If you prefer to work with **pen and pencil**:
 - Download and/or print the homework assignment.
 - Scan your completed work and combine all pages into a **single PDF file**.
 - Upload the PDF to the corresponding homework assignment on Canvas.
- If you prefer to use a **writing device** or type your solutions, you may complete the homework electronically and submit your final work directly as a **PDF**.
- You may also need to submit an **R file** if the homework includes a programming task.
- Your work must be **clear, organized, and legible**. Please follow these presentation guidelines:
 - Clearly circle or otherwise mark your final answers so they can be easily located on the page.
 - Show all necessary work for **nontrivial calculations**. Answers without sufficient supporting work may receive little or no credit.
- Name your submitted PDF using the following format:
`firstname_lastname_HW#.pdf`
For example: `john_smith_HW3.pdf`

Problem 3.1

The table below gives the distribution of grades for a previously taught STAT 312 class, where $X = 0$ represents an F, $X = 1$ represents a D, $X = 2$ represents a C, $X = 3$ represents a B, and $X = 4$ represents an A.

x	0	1	2	3	4
$P(X = x)$	0.01	0.02	0.10	0.30	0.57

- (a) Find the probability that a student in this STAT 312 class received a **C or better**.

- (b) Find the cumulative distribution function (CDF) of X ,

$$F(x) = P(X \leq x).$$

Express $F(x)$ as a piecewise function for all possible values of x . You do not need to graph the CDF.

Problem 3.2

There is an unfair die with unusual numbers on its six faces. Let Y denote the outcome of one roll of the die. The probability distribution of Y is given below.

Y	$P(Y = y)$
4	0.30
6	0.10
9	0.10
11	0.10
15	0.20
30	0.20

Suppose that every time you roll this die, you win a number of dollars equal to the number that appears. There is a catch, however: you must pay **\$12 for each roll**.

In the long run, do you expect to **win money**, **lose money**, or **break even**? Why? If you perform any calculations to answer this question, show your work.

Problem 3.3

A pile of 33 playing cards contains 5 spades, 7 clubs, 10 diamonds, and 11 hearts. Identify each of the following random variables as **Binomial**, **Geometric**, or **neither**.

- Draw cards from the pile one at a time at random **with replacement** until you see the first diamond. Let X = number of draws needed.
- Draw 5 cards all at once at random **without replacement**. Let X = number of clubs observed.
- Draw 7 cards one at a time at random **with replacement**. Let X = number of hearts observed.

Problem 3.4

A food corporation is introducing a new brand of cereal. To promote its new cereal, 37% of boxes found at grocery stores will have a free toy inside. The probability that any particular cereal box contains a toy is 0.37. Assume that consecutive cereal boxes are independent of each other.

- (a) If you purchase five boxes of cereal, what is the probability that **at least one** of them will contain a free toy?

Let Y_1 = number of toys found in a sample of 25 cereal boxes.

- (b) What kind of discrete random variable is Y_1 ? Include the value of its parameter in your answer, as we did in the class.
- (c) What are the possible values of Y_1 ?
- (d) What is the probability that none of the cereal boxes will contain a free toy?
- (e) What is the probability that exactly 8 of the cereal boxes will contain a free toy?
- (f) What is the probability that exactly 9 of the cereal boxes will contain a free toy?

Let Y_2 = number of boxes required to find the first free toy.

- (g) What kind of discrete random variable is Y_2 ? Include the value of its parameter in your answer, as we did in the class.
- (h) What are the possible values of Y_2 ?
- (i) What is the probability that $Y_2 = 4$?

Problem 3.5

A cashier works the midnight shift at a grocery store. The number of customers who come to her register per hour follows a Poisson distribution with $\lambda = 5.4$.

Let $Y_3 =$ **number of customers who come to her register per hour**.

(a) What is the probability that exactly 3 customers will come to her register in the next hour?

(b) What is the probability that four or more customers will come to her register in the next hour?

Let $Y_4 =$ **waiting time (in hours) until the next customer comes to her register**.

(c) What kind of continuous random variable is Y_4 ? Include the parameter value in your answer.

(d) On average, what is the waiting time (in hours) until the next customer comes to her register?

(e) The cashier would like to sneak outside for a cigarette break during her shift. If her cigarette break will last 14 minutes, what is the probability that no customers will come to her register while she is gone?

Problem 3.6

You and your family move to a beautiful small town named Disasterberg at the base of a mountain. Scientists have warned the public that the mountain is an active volcano and erupts, on average, once every 522 years. If the volcano were to erupt, the town would be destroyed. Assume that the time between eruptions follows an exponential distribution. Let $\mathbf{Y_5}$ = **waiting time (in years) until the next eruption.**

- (a) What is the probability that the volcano will erupt within the next 30 years?
- (b) What is the probability that you and your descendants will be able to live in the town peacefully, with no eruptions, for the next 100 years?
- (c) If $P(\text{no eruptions for the next } k \text{ years}) = 0.90$, find k .

Problem 3.7

A scientist studying flu epidemics claims that the waiting time, in years, until the next flu outbreak in Disasterberg follows a uniform distribution on the interval $(0, 11)$. If the scientist is correct, what is the probability that the town will not suffer any flu outbreaks for the next 6 years?